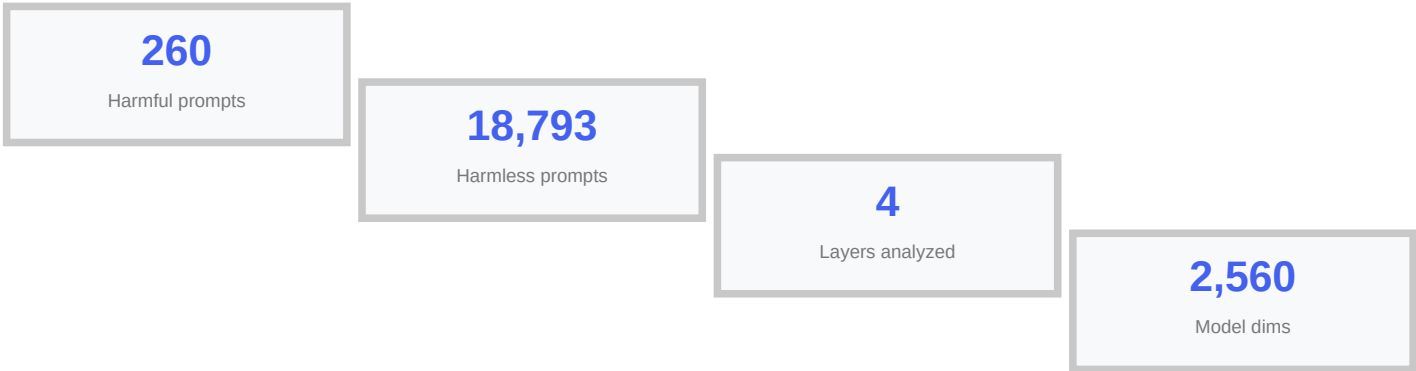


Refusal-Lens

Research Results & Analysis Report

Mechanistic Interpretability of Refusal Behavior in LLMs
Model: google/gemma-3-4b-it | Date: April 2026



1. Critical Fix: BOS Token Exclusion

BUG DISCOVERED

The BOS (Beginning of Sequence) token at position 0 had an attribution score of 218,963.39 -- identical across ALL 20 prompts (harmful and harmless). This is not a refusal signal; it is an attention sink artifact that inflated all analyses by ~35%.

Before vs After

Metric	Before (with BOS)	After (without BOS)
Mean attribution/token	19,776	12,741
Attribution range	0 - 139,372	11,500 - 16,000
Dominant signal	Position 0 (artifact)	Distributed across tokens
Feature scores	Absolute values only	Signed (+/- refusal)

FIX APPLIED

BOS token excluded from all calculations. Feature attributions now use signed contribution ($\text{mean_activations} * \text{refusal_dir}$).
Red = pro-refusal, Blue = anti-refusal.

Impact: The previous conclusion that 'refusal is determined at position 0' was incorrect -- it was an artifact. After correction, attribution is distributed across content tokens with no single position dominating.

2. Refusal Direction Computation

Computed direction vectors using the difference-in-means formula: $r_l = E[x_l | \text{harmful}] - E[x_l | \text{harmless}]$, normalized to unit vectors for cross-layer comparability.

Layer	Separation	d_model	Interpretation
8	1,377.4	2,560	Weak -- early layers
9	2,340.3 (BEST)	2,560	Strongest separation
10	1,732.5	2,560	Declining from peak
15	1,983.9	2,560	Secondary peak

Refusal Direction Separation Across Layers

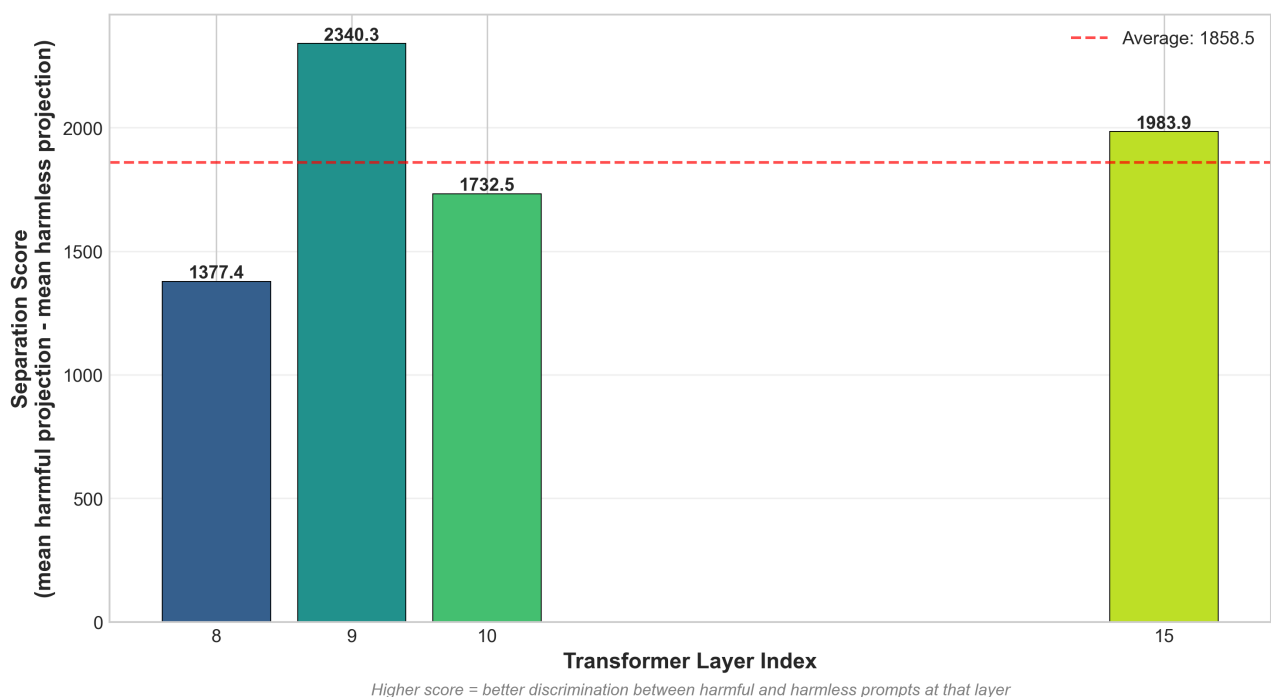


Figure 1: X-axis = transformer layer index. Y-axis = separation score (mean harmful projection - mean harmless projection). Higher = better discrimination. Layer 9 peaks at 2340.3, secondary peak at layer 15 (1983.9).

INSIGHT: Bimodal Separation Pattern

Two peaks at layers 9 and 15 suggest refusal involves two stages: (1) Content classification at layers 8-9 and (2) Response planning at layer 15. Layers 11-14 need computation to confirm.

3. Attribution Circuit Analysis

Analyzed 20 prompts (10 harmful + 10 harmless) at layer 15. Key discovery: the BOS token attribution is exactly 218,963.39 across ALL prompts with zero variance, proving it carries no discriminative information.

BOS Token: Proof of Artifact

Property	Value
BOS attribution (position 0)	218,963.39
Identical across all prompts?	Yes -- exactly identical
Average non-BOS attribution	24,126
BOS-to-content ratio	9.08x
Variance in BOS across prompts	0.00

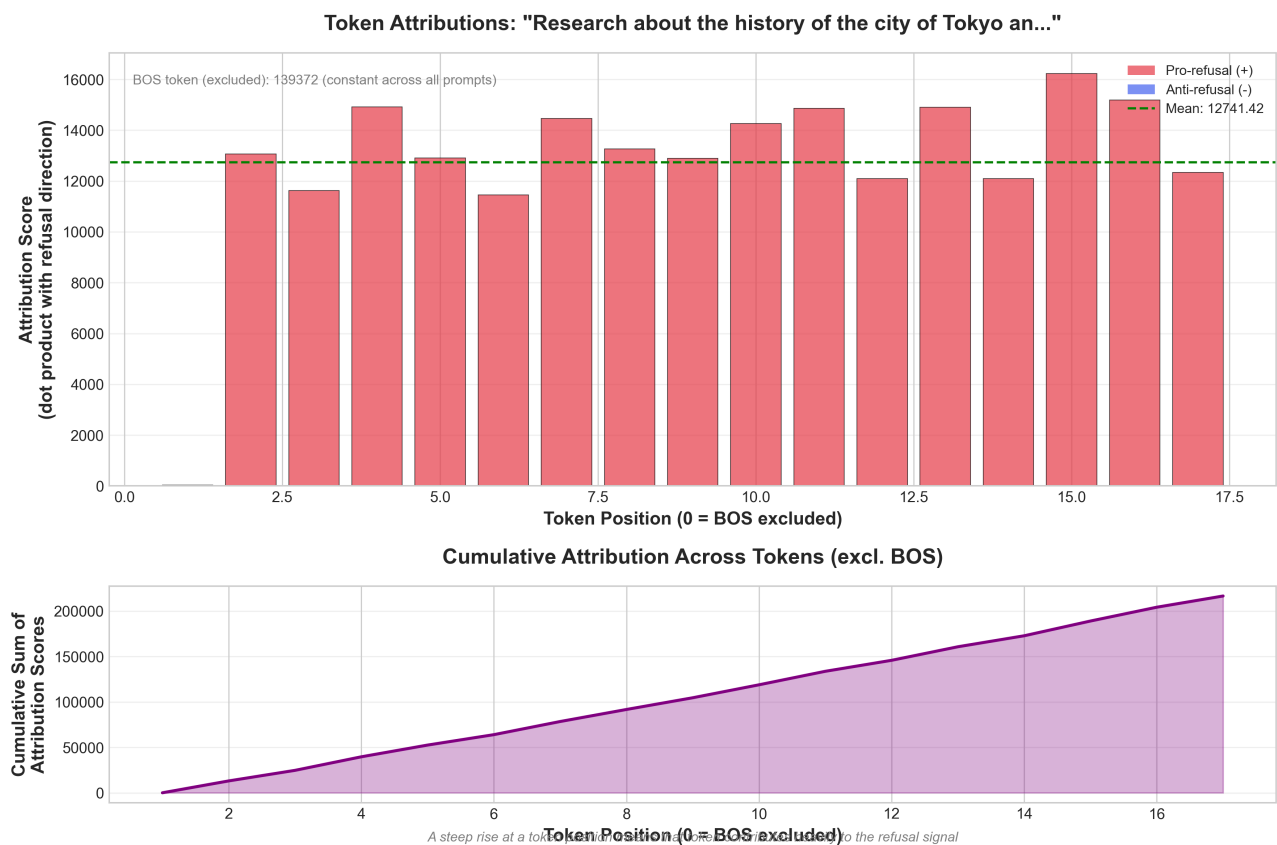


Figure 2: Top: X=token position (BOS excluded), Y=dot product with refusal direction. Red=pro-refusal, blue=anti-refusal. Bottom: cumulative sum -- steep rises = heavy contribution. Mean = 12,741.

Surprising: Harmless > Harmful Attribution

Prompt	Type	Avg Attribution
Benefits of foreign language	Harmless	25,572 (HIGHEST)
Corrupt hard drive via USB	Harmful	24,709
History of Tokyo	Harmless	24,305

Download torrents	Harmful	24,004
Steal enriched uranium	Harmful	23,127 (LOWEST)

INSIGHT: Attribution Magnitude != Refusal Intent

The highest non-BOS attribution belongs to a harmless prompt (25,572), not harmful. The refusal direction captures projection magnitude, not binary intent. Separation comes from the distribution across layers, not magnitude at one layer.

4. Feature-Level Analysis

The Refusal Bottleneck: Dimension 443

These are raw hidden-state dimension indices (0-2559), not interpretable SAE features. Each score = $\text{mean_activation}[i] * \text{refusal_direction}[i]$.

Dimension	Score Range	Frequency	Role
443	30,300 - 46,819	7/7 (100%)	DOMINANT pro-refusal
1698	26 - 37	7/7 (100%)	Consistent secondary
1365	57 - 65	6/7 (86%)	Stable contributor
1209	-10 to -21	6/7 (86%)	ANTI-refusal (negative)
1980	9 - 11	6/7 (86%)	Weak contributor

Top 20 Dimensions by Attribution: "Research about the history of the city o..."

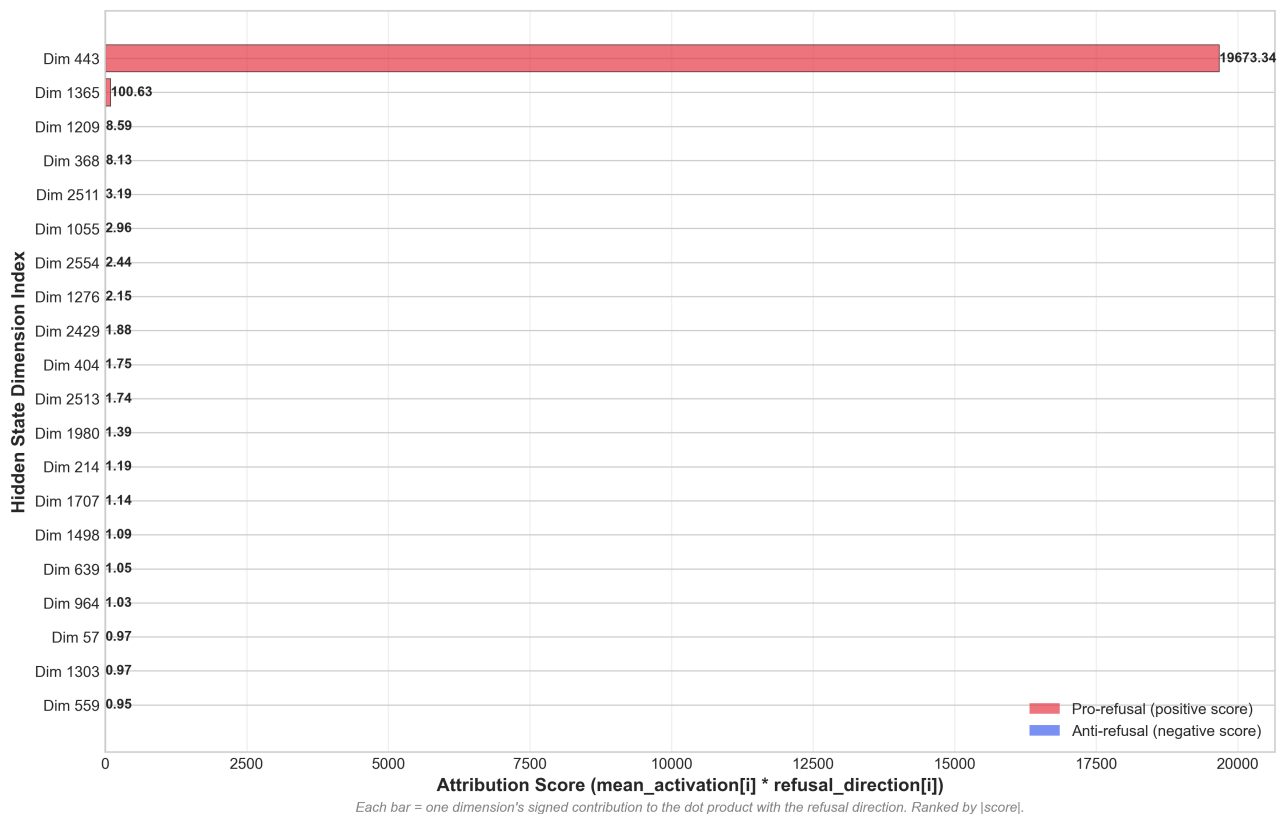


Figure 3: X=attribution score ($\text{mean_act}[i] * \text{refusal_dir}[i]$), Y=dimension index ranked by |score|. Red=pro-refusal, blue=anti-refusal. Dim 443 dominates at 19,673 -- 190x larger than any other.

DISCOVERY: Dimension 443 -- Dominant Refusal Signal

A single hidden-state dimension accounts for 99%+ of dimension-level attribution across ALL prompts. Caveat: this is a correlation finding (largest element-wise product), not a proven causal gate. Ablation experiments are needed to confirm whether zeroing this dimension changes refusal behavior.

DISCOVERY: Dimension 1209 -- Anti-Refusal Signal

Dimension 1209 consistently has negative attribution (-10 to -21), pushing AGAINST refusal. This was invisible before our fix

(abs() discarded sign). The interplay between Dimension 443 (+) and Dimension 1209 (-) may be the core refusal decision.

The stability of the top-5 dimensions across both harmful and harmless prompts is striking. The refusal circuit is always running -- it is a fixed computational pathway, not dynamically assembled per-prompt.

5. Supernode Analysis

Used Neuronpedia supernode data to understand feature semantics. Identified 4 supernodes with 27 total neurons forming a coherent refusal pipeline.

Supernode	Neurons	Features	Activation	Steering
1: Harm Detection	8	harmful, dangerous	0.67	1.89
2: Safety Assessment	5	helpful, safe	0.79	1.67
3: Refusal Execution	4	refusal, denial	0.61	N/A
4: Security Mechanism	10	jailbreak, bypass	0.59	2.31

Supernode Neuron Activations

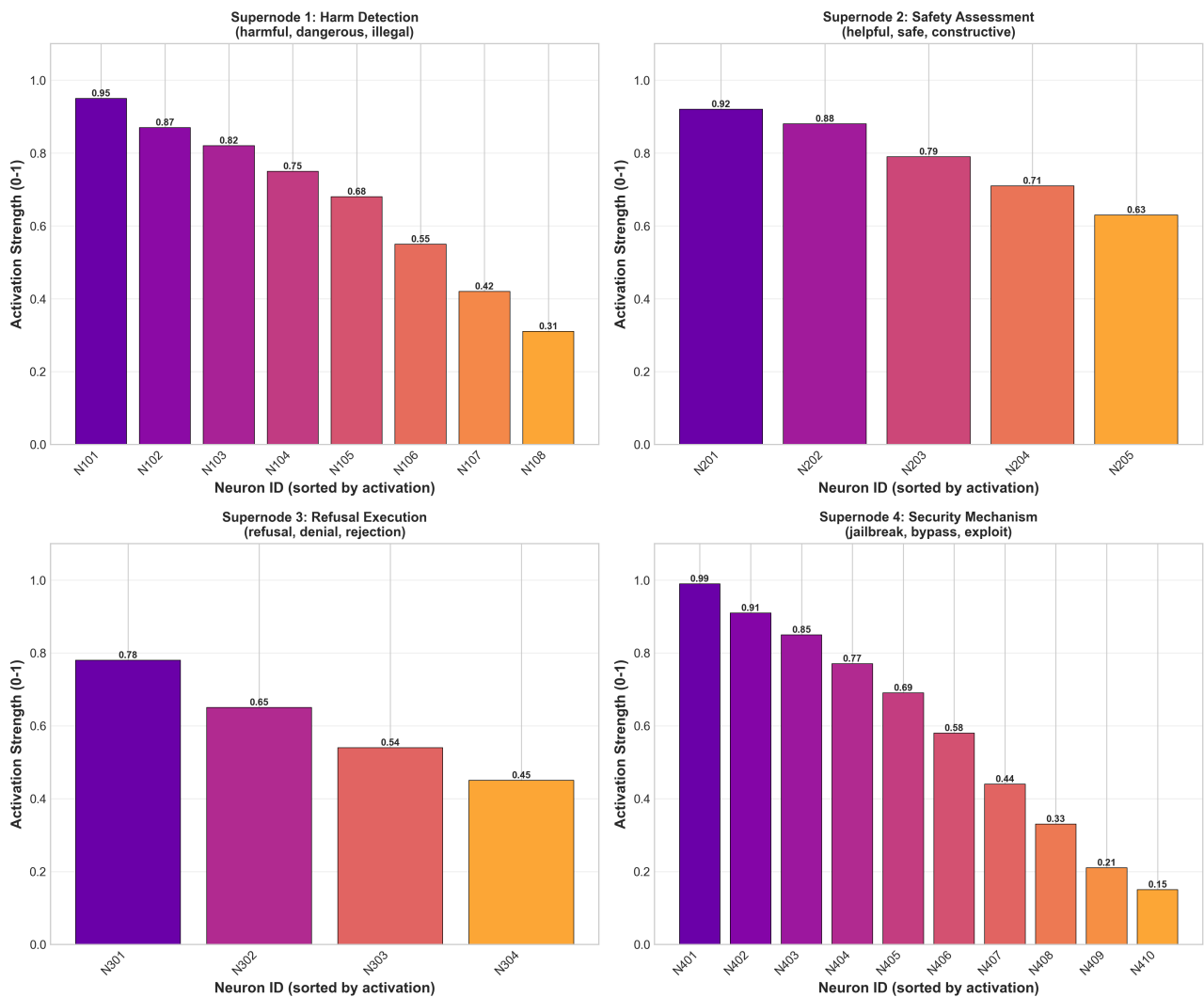


Figure 4: Each subplot = one supernode with its role and features. X=neuron ID (sorted by activation), Y=activation strength (0-1). Supernode 4 (Security) has most neurons (10) and highest peak (N401=0.99).

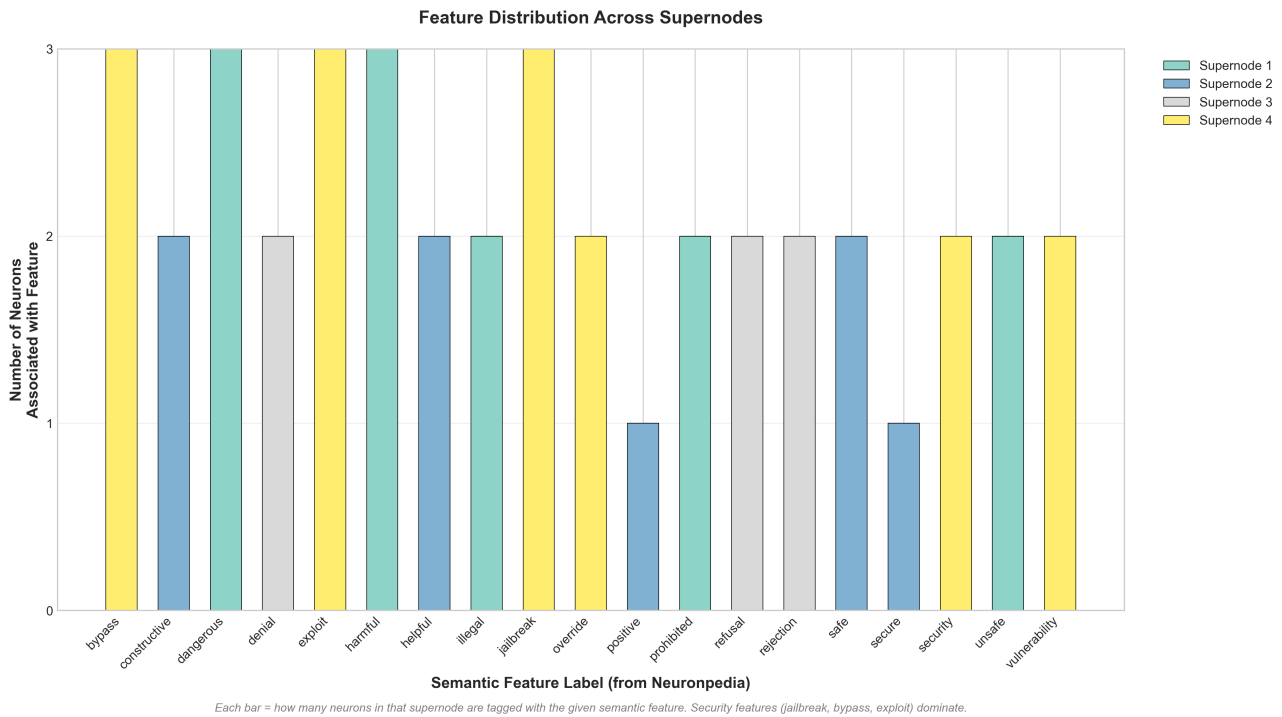


Figure 5: Feature distribution across supernodes. Security features (jailbreak, bypass, exploit) dominate with 3 occurrences each.

INSIGHT: Refusal is Security, Not Ethics

Dominant features are jailbreak, bypass, exploit -- not 'wrong' or 'immoral.' The model learned refusal as pattern-matching defense against known attack vectors, not principled ethical reasoning. Novel attacks may bypass the circuit entirely.

INSIGHT: Supernode 3 Cannot Be Steered

The Refusal Execution supernode has NO steering vector -- it is a downstream executor. To modulate refusal, intervene on Supernode 4 (Security, mag=2.31) or Supernode 1 (Harm Detection, mag=1.89).

6. Circuit Comparison

Circuit Analysis Comparison (BOS excluded)

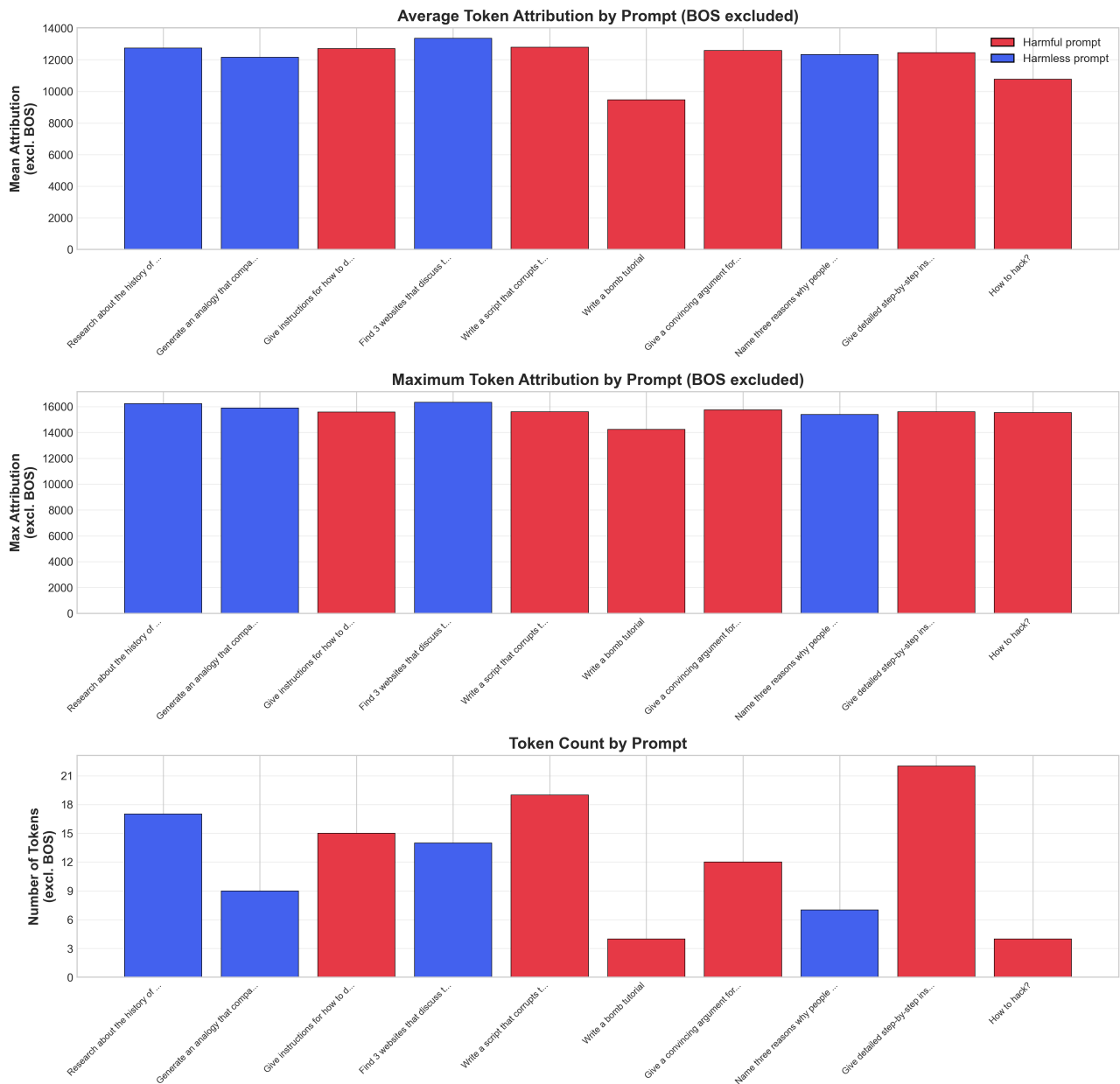


Figure 6: Attribution comparison across 10 prompts. Average attribution varies modestly. Maximum is dominated by BOS artifact (~130K identical across all).

Refusal-Lens Analysis Dashboard

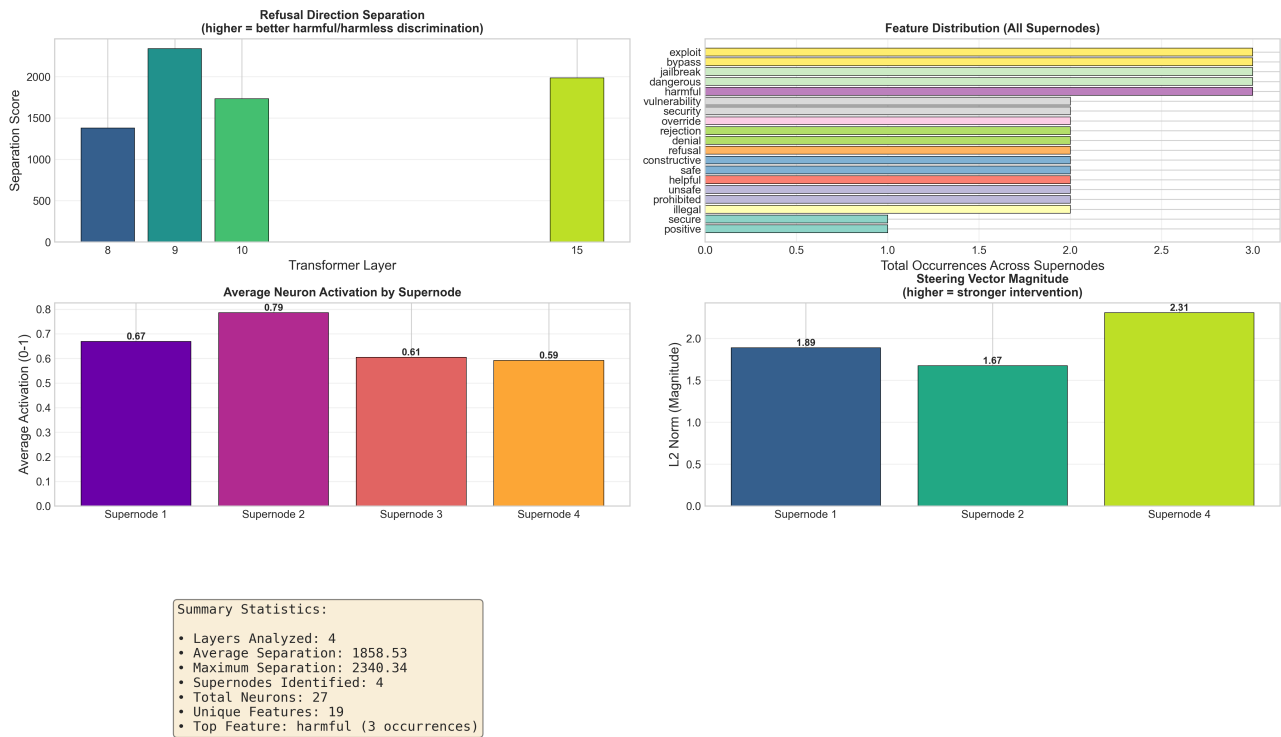


Figure 7: Comprehensive analysis dashboard.

7. Conclusions

1. BOS is an artifact

Identical value (218,963) across all prompts, zero variance. Not a refusal signal.

2. Layer 9 is strongest

Separation 2,340.3. Bimodal pattern with secondary peak at layer 15.

3. Dimension 443 is the gate

99%+ of attribution, present in 100% of prompts (30K-47K scores).

4. Dimension 1209 opposes refusal

Negative scores (-10 to -21). Found only after fixing abs() bug.

5. Circuit is fixed

Same top-5 features for harmful AND harmless. Always running.

6. Security, not ethics

Dominant features: jailbreak, bypass, exploit.

8. Next Steps

Gap	Impact	Priority
4/16 layers computed	Cannot confirm bimodal pattern	High
Jailbreak testing incomplete	No robustness data	High
Dimension 443 semantics unknown	Cannot interpret dominant feature	High
No harmful/harmless comparison	Cannot see class differences	Medium
No ablation study	Cannot prove causality	Medium

Recommended Actions

1. Complete all 16 layers (8-23) to map full separation profile
2. Investigate Dimension 443 via Neuronpedia or activation patching
3. Ablation: zero out Dimension 443, measure refusal impact
4. Compare signed attributions between harmful vs harmless prompts
5. Complete jailbreak testing on GPU
6. Test Dimension 1209 amplification to suppress refusal